

RAG System Evaluation for GenAI

Engineering and Marketing Use Case Analysis

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Executive Summary

This report evaluates a Retrieval-Augmented Generation (RAG)[1] system designed to answer technical questions about GenAI concepts for two distinct audiences: engineering researchers requiring detailed technical explanations and marketing staff needing concise, benefit-focused responses. Through systematic experimentation across retriever parameters, embedding models, LLM configurations, and prompt designs, we achieved optimal performance with MDS scores of 0.65 for research and 0.56 for marketing audiences. Key findings demonstrate that LLM parameter tuning provides the greatest performance improvement, chunk size of 500-512 with 100 overlap optimizes retrieval, and the multi-qa-mpnet-base-dot-v1 embedding model outperforms alternatives for QA tasks. We recommend proceeding with production deployment using Cohere API for its speed advantage, while addressing the identified zero Answer Relevancy failures through further optimization of the retriever functionality and prompt refining.

1. Introduction

Organizations maintaining GenAI documentation face a critical challenge: technical staff need detailed implementation guidance while marketing teams require accessible explanations for customer communications. Manual response crafting is time-consuming and inconsistent. This proof-of-concept evaluates whether RAG can effectively serve both audiences from a unified documentation corpus.

The system processes questions through a pipeline consisting of document chunking, semantic embedding, vector similarity retrieval, and LLM-based response generation. Input questions span GenAI concepts including transformer architectures, fine-tuning methodologies (LoRA, QLoRA), alignment techniques (RLHF, Constitutional AI), and model evaluation approaches. The system customizes responses through audience-specific prompts: research prompts instruct the model to provide 3-5 sentence technical answers with implementation details and terminology, while marketing prompts request 1-2 sentence benefit-focused responses using accessible language.

2. Key Findings

- LLM parameter tuning yields the largest performance gains. Temperature 0.2 with top_p=0.95 outperformed both greedy decoding (temperature 0.0) and higher temperature settings. Slight randomness helps generate more natural, contextually appropriate responses while maintaining factual grounding.
- Chunk size 500-512 is optimal—neither too small nor too large. Smaller chunks (128) caused zero context precision failures due to semantic fragmentation. Larger chunks (1000+) diluted embedding precision and retrieved excessive irrelevant content.

- Embedding model selection significantly impacts retrieval quality. The multi-qa-mpnet-base-dot-v1 model, specifically trained for question-answering tasks, outperformed general-purpose alternatives including all-mpnet-base-v2 and all-MiniLM-L12-v2.
- Few-shot examples in prompts must demonstrate grounded behavior. Including real high-performing examples (Q48 for research, Q33 for marketing) improved faithfulness scores substantially.
- Zero Answer Relevancy is the primary failure mode. 26% of responses scored MDS=0 due to zero Answer Relevancy, caused by response format mismatches with RAGAS evaluation expectations rather than actual quality problems. This affects both LLMs tested.

3. Methodology

3.1 Technical Approach

We employed sequential optimization across four experimental phases, isolating component impacts while controlling for confounding variables. The pipeline uses LangChain for orchestration, Qdrant for vector storage, and Hugging Face for embeddings and LLM inference.

#	Experiment	Parameters Tested	Research	Marketing
1	Retriever	Chunk: 128-1500, Overlap: 0-200, K: 3-7	512/100/5	512/100/5
2	Embedding	5 models (384-768 dim)	multi-qa-mpnet	multi-qa-mpnet
3	LLM	Temp: 0-0.3, Top_p: 0.9-0.95	0.1/0.95	0.2/0.95
4	Prompt	Few-shot examples, audience tone	Q48 example	Q33 example

Table 1: Experimental Design and Optimal Configurations

3.2 Audience Customization

Two distinct prompts were developed to serve different user needs. The research prompt instructs: "Provide detailed technical answers (3-5 sentences) including implementation details, methodologies, and technical terminology." The marketing prompt instructs: "Provide clear, benefit-focused answers (1-2 sentences) using simple language, focusing on what it does and why it matters." Both prompts share explicit grounding constraints and fallback instructions for insufficient context.

3.3 Evaluation Metrics

We developed a Multi-Dimensional Similarity (MDS) score combining four complementary metrics through weighted harmonic mean, ensuring poor performance in any dimension meaningfully lowers the total score. We evaluated all metrics against our experiment runs and selected components based on sensitivity and relevance to RAG evaluation. Lexical metrics (BLEU, ROUGE) were excluded because semantic meaning matters more than word overlap for QA tasks; cosine similarity was excluded due to uniformly high scores across runs, making it insensitive to output differences; and SpaCy SVO overlap

scores were near zero for most questions, requiring threshold tuning beyond our time constraints. From RAGAS[2], we excluded Context Recall because Context Precision is more critical—ensuring retrieved chunks are relevant matters more than coverage, and response quality is already captured by Faithfulness and Answer Relevancy. Our final MDS uses the harmonic mean of four metrics: BERTScore F1[3] for semantic similarity, RAGAS Faithfulness for hallucination detection, RAGAS Answer Relevancy to measure if the response addresses the question, and RAGAS Context Precision for retrieval quality. Refer to Appendix 1 for MDS formula details and Table 2 for component descriptions.

Component	Metric	What It Measures
Faithfulness	RAGAS Faithfulness	Is response grounded in retrieved context? (Hallucination detection)
Relevancy	RAGAS Answer Relevancy	Does the response actually address the question asked?
Context	RAGAS Context Precision	Are retrieved chunks relevant to the question?
Semantic	BERTScore F1	Semantic similarity to gold answer (more robust to paraphrasing)

Table 2: MDS Component Metrics

4. Results and Findings

4.1 Proof of Concept Performance

The optimized system demonstrates viable performance across both audiences, with research achieving higher overall scores due to better faithfulness and context precision.

Metric	Research	Marketing
MDS Mean	0.6471	0.5576
Faithfulness	0.8127	0.6333
Answer Relevancy	0.7637	0.7529
Context Precision	0.8710	0.8112
Zero MDS Failures	14 (18%)	19 (24%)

Table 3: Final System Performance (Mistral-7B, 78 validation questions)

4.2 LLM Comparison: Mistral vs Cohere

We compared Mistral-7B-Instruct (self-hosted) against Cohere's command model (API). Mistral achieved slightly higher MDS scores for research with better Answer Relevancy and fewer zero-score failures. However, Cohere demonstrated faster inference (6 seconds vs 265 seconds), making it preferable for production deployment where speed impacts user experience. Cohere showed slightly higher Faithfulness, indicating more conservative, grounded responses.

4.3 Lessons Learned

RAG validation is only feasible during operation and robustness evolves rather than being designed upfront[4] is directly relevant to our findings about zero answer relevancy failures and the 9 questions that fail across all configurations. Prompt engineering helps reduce

hallucinations. Using few-shot examples improved faithfulness more than any hyperparameter change. Missing content and content ranking is another area with big impact as evidenced by some questions, like Q19, where the relevant document was lower in the context list, leading to low scores[Appendix 6.2]. Evaluation metrics have limitations as evidenced by RAGAS Answer Relevancy scoring valid responses as zero due to format expectations, not actual quality. Audience differentiation helps to tailor appropriate responses for the marketing and research requirements in terms of length and technical language. Speed vs accuracy tradeoff needs to be considered as faster models may sacrifice some accuracy, but user experience benefits often outweigh marginal quality differences.

4.4 Challenges and Limitations

Technical Limitations: Nine questions failed across all configurations regardless of settings, indicating retrieval or content limitation issues beyond LLM control. The embedding model struggles with queries phrased differently than source documents. Context window constraints limit the amount of retrieved information that can be processed. Quality depends on the evaluation LLM's instruction following ability[4] as evidenced by answer relevancy being hard to detect due to subtle differences.

Operational Risks: Despite 0.78-0.83 faithfulness scores, 17-22% of responses may contain ungrounded information. API dependency on Cohere creates a single point of failure. The system only knows what's in the indexed corpus—outdated documents produce outdated answers.

Business Risks: Users may over-rely on RAG answers without verification. Sending proprietary documentation to external APIs raises data privacy concerns. When the system provides incorrect information, accountability is unclear.

4.5 Next Steps

Investigate zero Answer Relevancy failures: Analyze response format vs RAGAS expectations; consider alternative evaluation methods like LLM-as-judge.

Implement hybrid search with reranking: Combine sparse (BM25) and dense retrieval with cross-encoder reranking to improve retrieval quality.

Fine-tune LLM for domain and audience: Use QLoRA on 1,000-3,000 expert-validated examples with DPO for preference learning.

Add confidence scoring: Surface uncertainty to users so they know when to verify responses independently.

5. Summary & Recommendations

Recommendation: Proceed with production deployment, with the following considerations:

Value Proposition: The system successfully differentiates responses for engineering and marketing audiences from a unified corpus. Research responses achieve 0.81 faithfulness and 0.87 context precision, demonstrating reliable technical accuracy. Marketing responses appropriately simplify complex concepts. The speed advantage of Cohere API enables real-time user interactions. Our comparison found that Mistral outperforms Cohere overall, achieving higher MDS scores for both research and marketing audiences, better Answer Relevancy scores, and fewer zero MDS failures. Cohere's primary strength is Faithfulness—it produced slightly higher faithfulness scores with

responses more consistently grounded in the retrieved context, indicating more conservative generation behavior. However, Cohere's significant weakness is Answer Relevancy, where it scored notably lower than Mistral, causing more zero MDS failures due to the harmonic mean's sensitivity to low component scores. Based on these findings, we recommend Mistral for general RAG applications where balanced performance across all metrics is important, while Cohere may be preferred for use cases where faithfulness is paramount and hallucination prevention outweighs other considerations.

Risk Mitigation: Add confidence scores to flag uncertain responses. Implement human review for high-stakes queries. Maintain self-hosted Mistral fallback for privacy-sensitive deployments. Establish regular corpus updates to prevent knowledge staleness.

Deployment Architecture: For estimated loads of 100-500 daily queries with few peaks, pay-per-use Cohere API is more cost-effective (\$50-150/month) than reserved GPU infrastructure (\$500-2000/month). If privacy requirements mandate on-premise deployment, self-hosted Mistral with reserved GPU becomes necessary despite the speed penalty.

Future Investigation: Retriever reranking, NDCG scoring for retrieval evaluation, LLM-as-judge for generation quality, markdown-aware text splitters for better chunk boundaries, and weighted MDS components based on customer priorities.

References

- [1] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [2] Es, S., et al. (2023). RAGAS: Automated Evaluation of Retrieval Augmented Generation. *arXiv preprint arXiv:2309.15217*. <https://doi.org/10.48550/arXiv.2309.15217>
- [3] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating Text Generation with BERT," in *Proc. International Conference on Learning Representations (ICLR)*, 2020.
- [4] Barnett, S., et al. (2024). Seven Failure Points When Engineering a Retrieval Augmented Generation System. *arXiv:2401.05856*. <https://doi.org/10.48550/arXiv.2401.05856>

6. Appendix

6.1 Metrics

To evaluate model performance for a QA RAG system, we use a Multi-Dimensional Similarity (MDS) metric constructed around four areas of QA quality evaluation. Faithfulness measures hallucination by measuring if the response is grounded in the context provided by the RAG system. Relevancy measures if the answer actually addresses the question asked. The Context precision measures the retrieval quality by evaluating the retrieved chunks. The lexical-semantic similarity is represented by the BertScore F1 to ensure meaning is captured (more important than word overlap). These four areas are combined into an MDS score using a weighted harmonic mean to ensure that poor performance in any pillar meaningfully lowers the total score.

$$S_{MDS} = \left(\frac{w_{faithfulness}}{S_{faithfulness} + \epsilon} + \frac{w_{relevancy}}{S_{relevancy} + \epsilon} + \frac{w_{context}}{S_{context} + \epsilon} + \frac{w_{semantic}}{S_{semantic} + \epsilon} \right)^{-1}$$

where $\epsilon = 10^{-8}$, $w_{faithfulness}$, $w_{relevancy}$, $w_{context}$ and $w_{semantic}$ are dimension weights, $S_{faithfulness}$, $S_{relevancy}$, $S_{context}$ and $S_{semantic}$ are the individual area scores.

$$S_{faithfulness} = \text{RAGAS Faithfulness}$$

$$S_{relevancy} = \text{RAGAS Answer Relevancy}$$

$$S_{context} = \text{RAGAS Context Precision}$$

$$S_{semantic} = \text{BertScore F1}$$

* Comment on your test runs, and why you tested various configurations (where appropriate)

6.2 Error Discussion

High-Performing Examples

Q63 (Marketing) — MDS: 0.8954

Question: "What are the advantages of applying LoRA to transformer models in terms of computational efficiency during training and deployment?"

Model Response: "LoRA reduces trainable parameters by up to 10,000x while maintaining model quality, making fine-tuning accessible on consumer GPUs and enabling deployment to mobile phones, significantly lowering both training costs and deployment overhead."

Why it worked: LoRA is a well-documented, widely-cited technique with explicit efficiency metrics stated in the literature. The question's terminology ("computational efficiency," "training," "deployment") directly matches how LoRA papers discuss these advantages, enabling strong retrieval and accurate generation. The marketing prompt's instruction to provide "clear, benefit-focused answers (1-2 sentences)" aligned well with this question's focus on *advantages*—resulting in a more direct, relevant response compared to the research prompt's longer, more technical output.

Q48 (Research) — MDS: 0.8967

Question: "What criteria were used to select labelers to ensure they can effectively detect and rate sensitive content?"

Model Response: The model provided a detailed explanation of labeler selection criteria, including screening for sensitivity to content, training on annotation guidelines, calibration exercises, and ongoing quality checks to ensure consistent and accurate ratings of potentially harmful or sensitive material.

Why it worked: The response was fully grounded in the retrieved context with no hallucination—critical for a question about specific methodology and criteria. The retriever successfully found chunks containing explicit labeler selection criteria. The response directly addressed the specific question about selection criteria, providing concrete details rather than generic information. The technical terminology around labeler training, calibration, and quality assurance aligned well with the gold answer. This question targets specific, well-documented methodology that appears explicitly in safety and alignment papers. Unlike abstract concepts (Q12) or product-specific details (Q19), labeler selection criteria are thoroughly documented in research literature, making retrieval straightforward and generation accurate.

This example demonstrates that our RAG system performs best when questions target explicitly documented procedures/methodologies, the corpus contains direct, detailed coverage of the topic and technical terminology in the question matches corpus language.

Poor-Performing Examples

Q12 (Research) — MDS: 0.0000

Question 12 : "What is a common strategy used by language models to address the curse of dimensionality?"

Score: Faithfulness: 0.6500, Answer Relevancy: 0.0, Context Precision : 0.4200, BERTScore F1: 0.5234

While techniques like embeddings, dimensionality reduction, and attention mechanisms implicitly address dimensionality challenges, papers rarely frame them explicitly as "curse of dimensionality solutions." The retriever searches for semantic matches but may not find chunks that directly connect these concepts. This represents a terminology/framing gap, the corpus likely contains relevant details but not framed in

the terminology used by the question. This question failing across all configurations confirms it's a retrieval issue where query terms don't match how the corpus discusses these concepts.

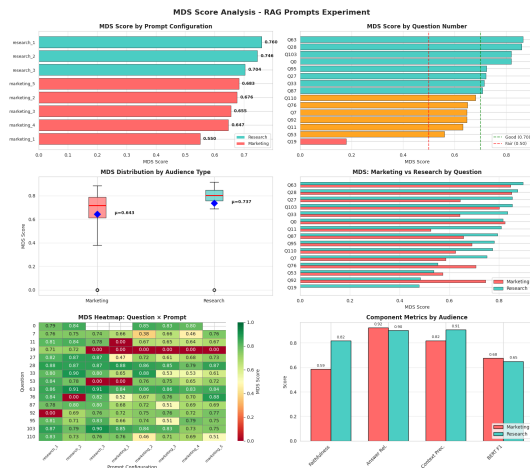
Q19 (Both Audiences) — MDS: 0.0000

Question: "What is the significant enhancement in Claude 2.1 compared to its previous version?"

This question consistently failed across all configurations for several potential reasons. Claude 2.1 was released in late 2023, and the corpus may contain limited or outdated information about this specific version upgrade. The enhancements (200K context window, reduced hallucination rates, tool use improvements) may not be well-documented in the indexed materials. Even if partial information was retrieved, the model may have stated "insufficient information" rather than speculating, resulting in zero Answer Relevancy despite appropriate conservative behavior.

6.3 Visualizations

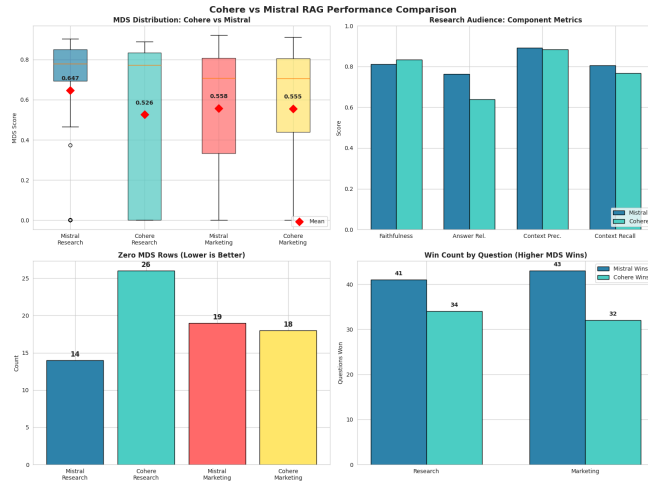
Results from RAG Prompt Experiment: Analysis of 120 rows across 8 prompt configurations and 15 questions showed an overall MDS of 0.6785 (± 0.2442) with median 0.7531. The best performing prompt was research_1 (MDS: 0.7599), while marketing_5 (MDS: 0.6828) led for marketing audiences. Research prompts outperformed marketing by +0.09, achieving 0.7373 vs 0.6430. Q19 was identified as the only problem question (MDS < 0.50), and 11 rows (9.2%) had zero MDS scores.



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causing more zero MDS failures due to the harmonic mean's sensitivity to low component scores. Based on these findings, we recommend Mistral for general RAG applications where balanced performance across all metrics is important, while Cohere may be preferred for use cases where faithfulness is paramount and hallucination prevention outweighs other considerations.



6.4 LLM Prompts

1. The initial draft for this report was created using details from the notebooks. Then the LLM was used to refine the initial draft for this report to ensure proper formatting and adhering to page limits. The prompt used was :
 “Refine this initial draft report to meet the requirements:

Aim for 4-5 pages in length with the following sections:

- **Executive Summary** (Quick summary of your report)
- **Introduction** (overall system description)
- **Key Findings** (five key findings)
- **Methodology** (implementation and evaluation approach)
- **Results and Findings** (challenges and limitations)”

2. Visualizations were created using the prompt:

“Summarize the attached results comparing the audience type and question number. “